



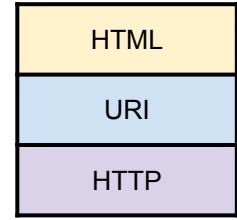
Hochschule
Bonn-Rhein-Sieg
University of Applied Sciences

Einführung in Web Engineering

CSS 1. Teil

Manfred Kaul

Diese 3 Konzepte sind die
Basis für das WWW:



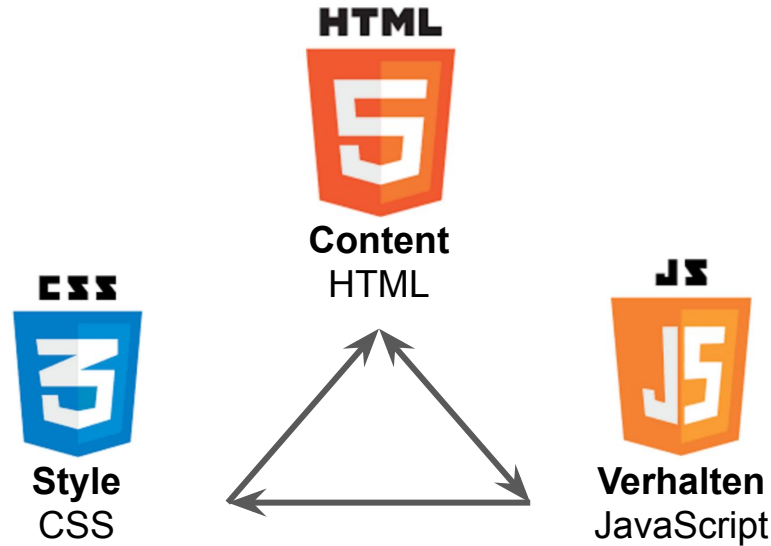
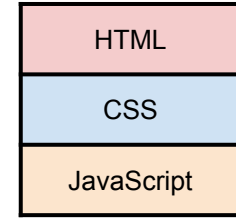
Bis hierher

die anfänglichen Ideen von Tim Berners-Lee.

Das erste Web war **sicher, performant, skalierbar, ...**

Was kam danach?

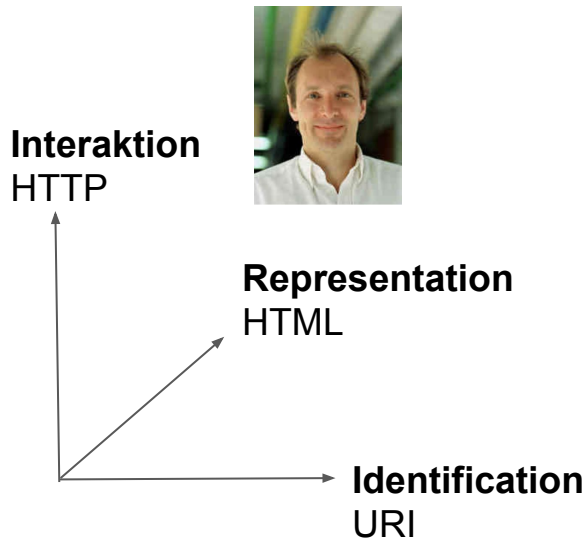
Heutige Browser-Triade



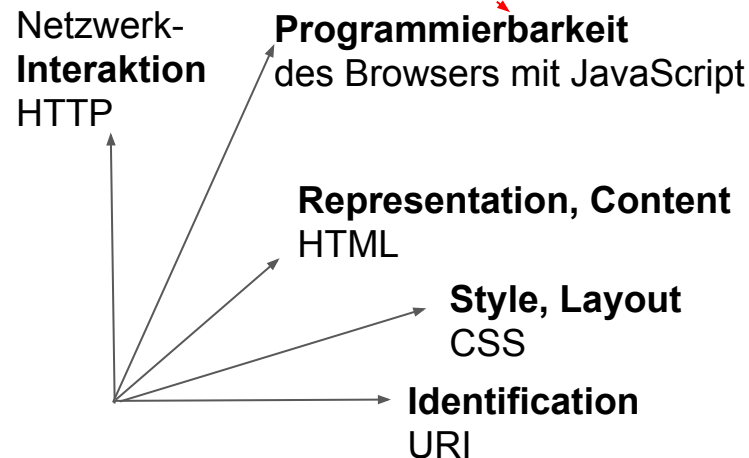
Weitere Dimensionen der Emergenz ...

HTML
CSS
JavaScript

Der Browser als
Code on Demand-
Plattform

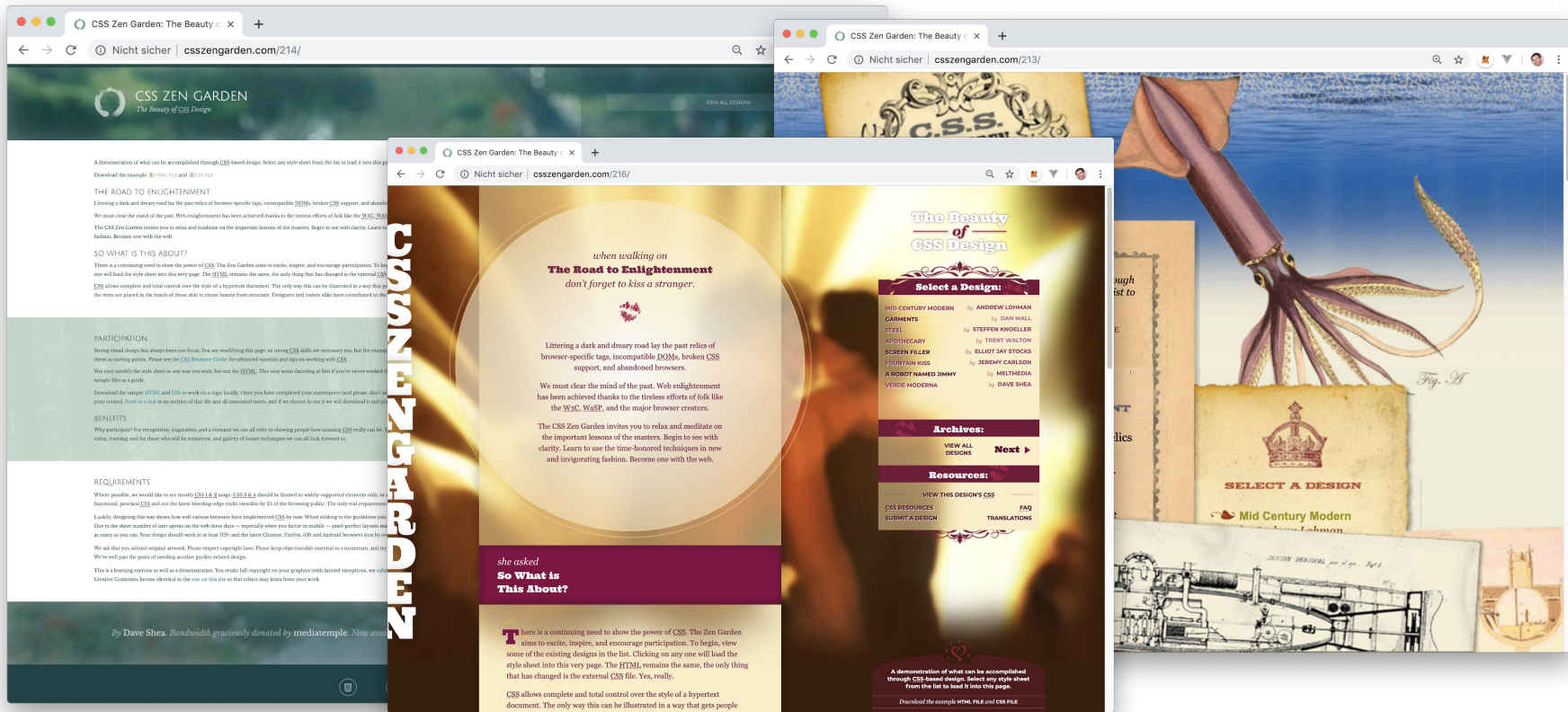


alte WWW-Triade

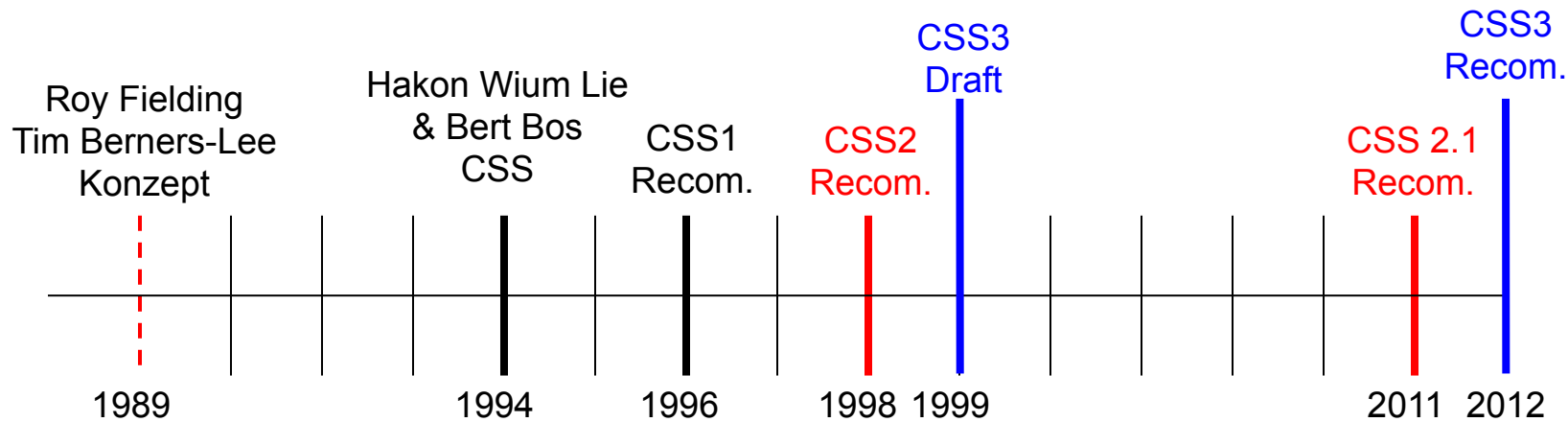


Das heutige WWW

Trennung von (strukturiertem) Inhalt und Layout



CSS Timeline



Was war denn hier passiert???

https://de.wikipedia.org/wiki/Cascading_Style_Sheets

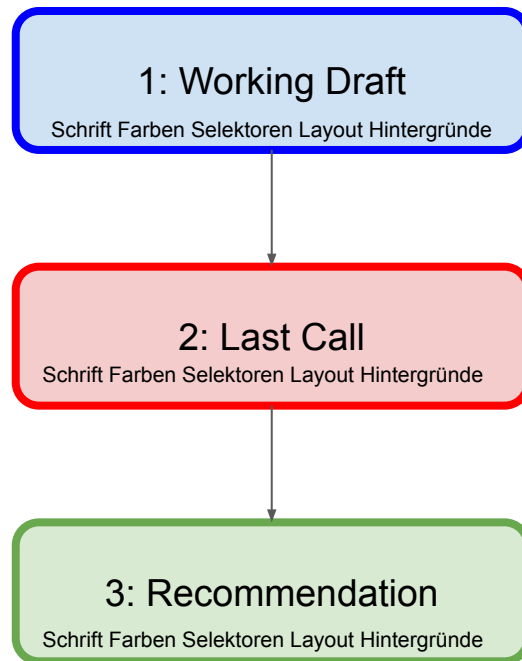
CSS 1 und 2 waren monolithische Standards

Alle CSS-Regeln zu Schrift, Farben, Selektoren, Layout, Hintergründe wurden in einem Standard *gemeinsam* verabschiedet

⇒ Alles musste fertig sein, damit der nächste Schritt möglich wurde.

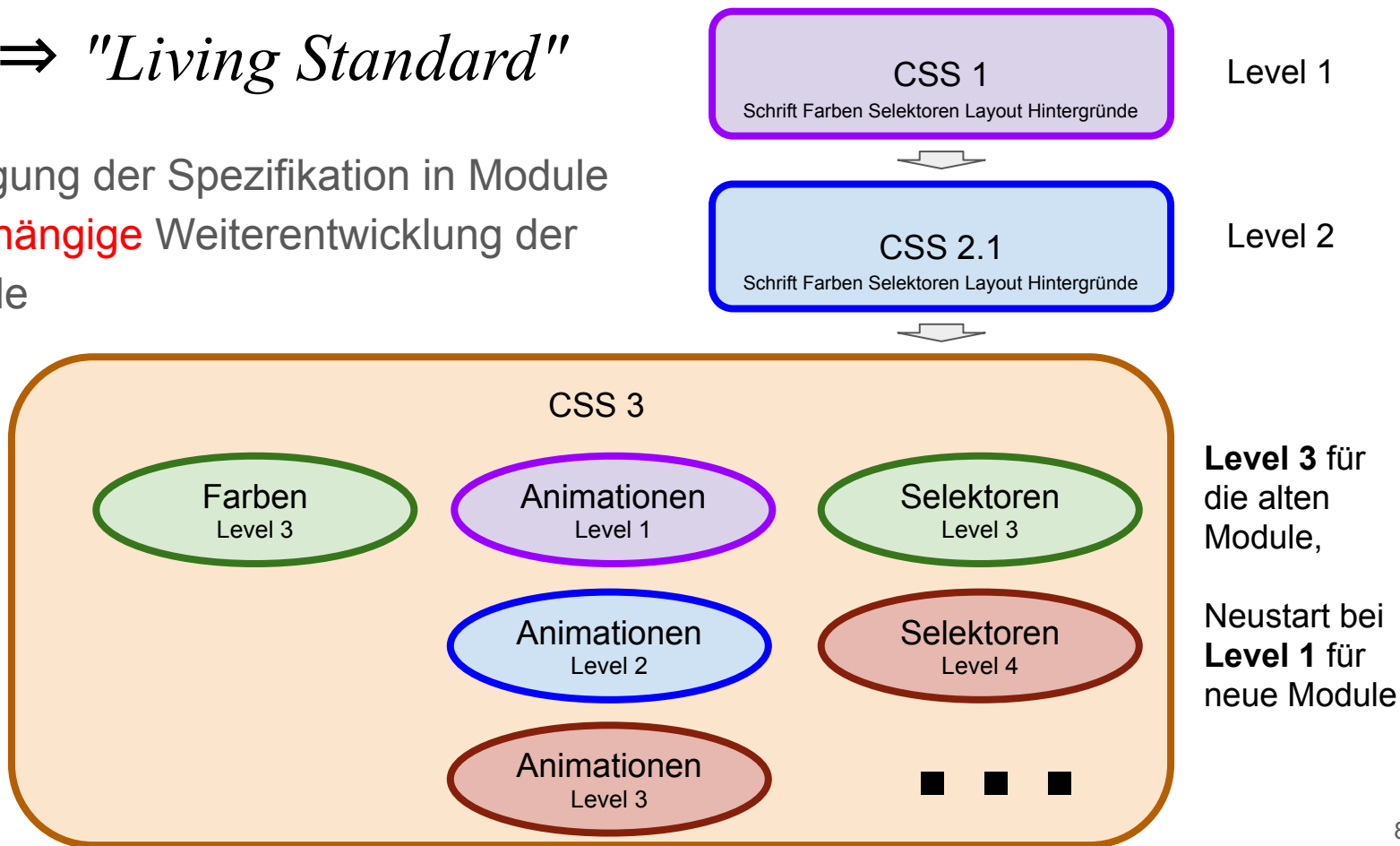
⇒ Verletzung des Prinzips der Orthogonalität verschiedener Entwicklungslinien

⇒ CSS-Standardisierung zu schwerfällig.

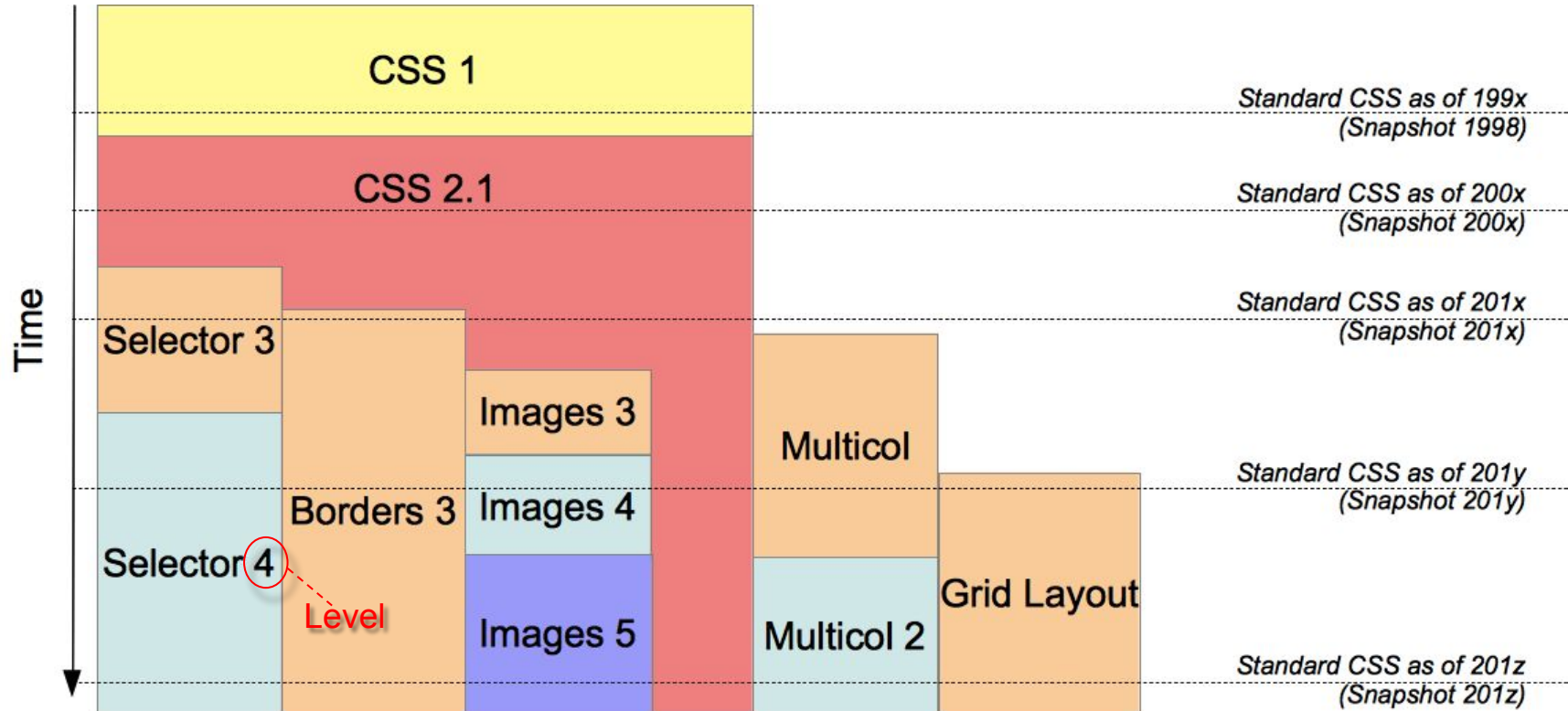


CSS3 $\Rightarrow$ "Living Standard"

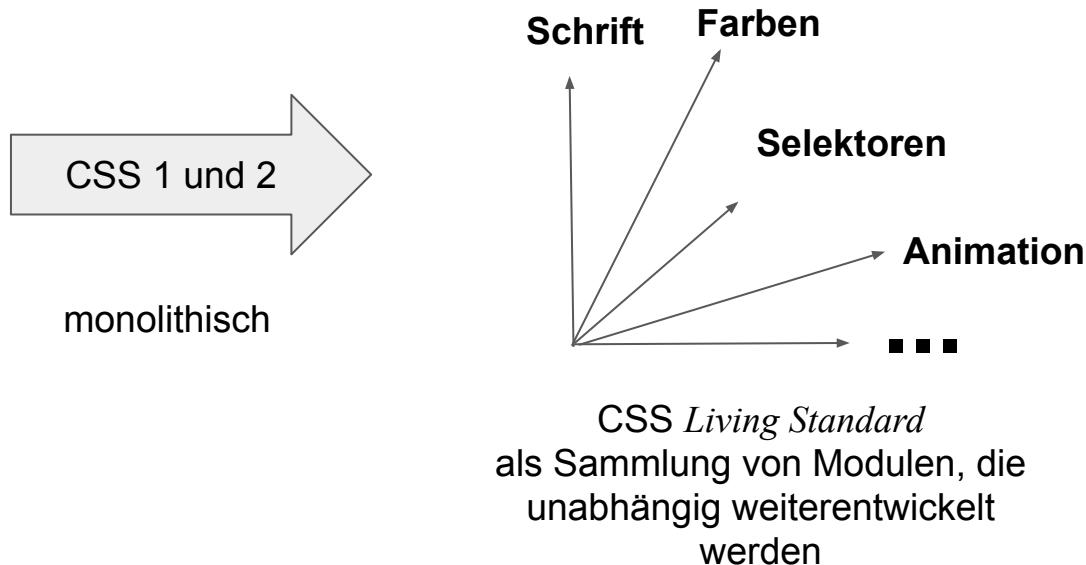
- Zerlegung der Spezifikation in Module
- **Unabhängige** Weiterentwicklung der Module



Zerlegung des CSS-Standards in Module



Seit das Prinzip der Orthogonalität in CSS angewendet wird, hat sich der Fortschritt wieder beschleunigt



TABELLARISCHE ÜBERSICHT DER SPEZIFIKATIONEN

Abgeschlossene Werke	Aktuell	Zukünftig		Mittlere Priorität	Aktuell	Zukünftig		
CSS Snapshot 2017	NOTE		Latest stable CSS	①	CSS Animations	WD	WD	①
CSS Snapshot 2015	NOTE			①	Web Animations 1.0	WD	WD	①
CSS Snapshot 2010	NOTE			①	CSS Text Level 3	LC	CR	①
CSS Snapshot 2007	NOTE			①	CSS Transforms	WD	WD	①
CSS Color Level 3	REC	REC		①	CSS Transitions	WD	CR	①
CSS Namespaces	REC	REC		①	CSS Cascading Variables	CR	PR	①
Selectors Level 3	REC	REC		①	CSS Box Alignment Module Level 3	WD	WD	①
CSS Level 2 Revision 1	REC	REC		①	CSS WIC Change Level 1	CR	PR	①
CSS Level 1	REC			①	Preview of CSS Level 2	FPWD	NOTE	①
CSS Print Profile	NOTE		Unmaintained, see Snapshot	①	Mittlere Priorität			
Media Queries	REC	REC		①	CSS Paged Media Level 3	WD	WD	①
CSS Style Attributes	REC	REC		①	CSOM View	WD	WD	①
				①	Selectors Level 4	WD	WD	①
				①	CSS Intrinsic & Extrinsic Sizing Module Level 3	WD	CR	①
					CSS Overflow Level 3	FPWD	WD	①
				①	CSS Display Level 3	WD	WD	①
				①	CSS Pseudo-Elements Module Level 4	FPWD	WD	①
				①	Mittlere Priorität			
				①	CSS Backgrounds and Borders Level 4	WD	WD	①
				①	CSS Device Adaptation	WD	WD	①
				①	CSS Exclusions	WD	WD	①
				①	Filter Effects	WD	WD	①
				①	CSS Generated Content for Paged Media	WD	WD	①
				①	CSS Page Flows	FPWD	WD	①
				①	CSS Template Layout	NOTE	NOTE	①
				①	CSS Line Grid	WD	WD	①
				①	CSS Lists Level 3	WD	WD	①
				①	CSS Positioned Layout Level 3	WD	WD	①
				①	CSS Regions	WD	WD	①
				①	CSS Ruby Level 1	WD	WD	①
				①	CSS Tables Level 3	FPWD	WD	①
				①	CSS Object Model	WD	WD	①
				①	CSS Font Loading	LC	WD	①
				①	CSS Scoping Level 1	FPWD	WD	①
				①	Media Queries level 4	WD	WD	①
				①	Non-element Selectors	FPWD	WD	①
				①	CSS Inline Layout Module Level 3	WD	WD	①
				①	Motion Path Module Level 1	FPWD	WD	①
				①	CSS Round Display Level 1	WD	WD	①
				①	CSS Basic User Interface Module Level 4	FPWD	WD	①
				①	CSS Text Module Level 4	FPWD	WD	①
				①	CSS Painting API Level 1	FPWD	WD	①
				①	CSS Properties and Values API Level 1	FPWD	WD	①

Browser-Abhängigkeiten:

Can I use column-break-inside?

CSS3 Multiple column layout - CR

Method of flowing information in multiple columns

Global
unprefixed:
Germany
unprefixed:

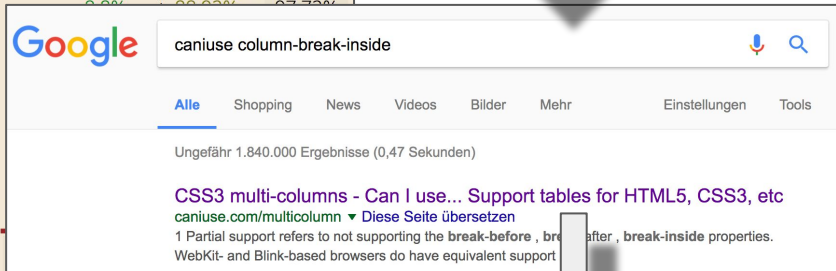
Current aligned Usage relative Date relative Show all

IE	Edge *	Firefox	Chrome	Safari	Opera	iOS Safari *
			1 2 49			
			1 55			
		1 51	1 56			1 9.3
	14	1 52	1 57	1 10	1 43	1 10.2
11	15	1 53	1 58	1 10.1	1 44	1 10.3
		1 54	1 59	1 TP	1 45	
		1 55	1 60	1 46		
		1 56	1 61			

Notes Known issues (9) Resources (10) Feedback

¹ Partial support refers to not supporting the **break-before**, **break-after**, **break-inside** properties. WebKit- and Blink-based browsers do have equivalent support for the non-standard **-webkit-column-break-*** properties to accomplish the same result (but only the **auto** and **always** values). Firefox does not support **break-***.

² Partial support refers to not supporting the **column-fill** property.



<http://caniuse.com/#feat=multicolumn>

Syntaktischer Aufbau von CSS-Regeln

Allgemein:

Selektor1 [, **Selektor2** [, ...]] {

Eigenschaft-1: Wert-1;

...

Eigenschaft-n: Wert-n[;]

$n \geq 0$

}

CSS-Deklaration

/ Kommentar */*

/ In eckigen Klammern stehen optionale Angaben */*

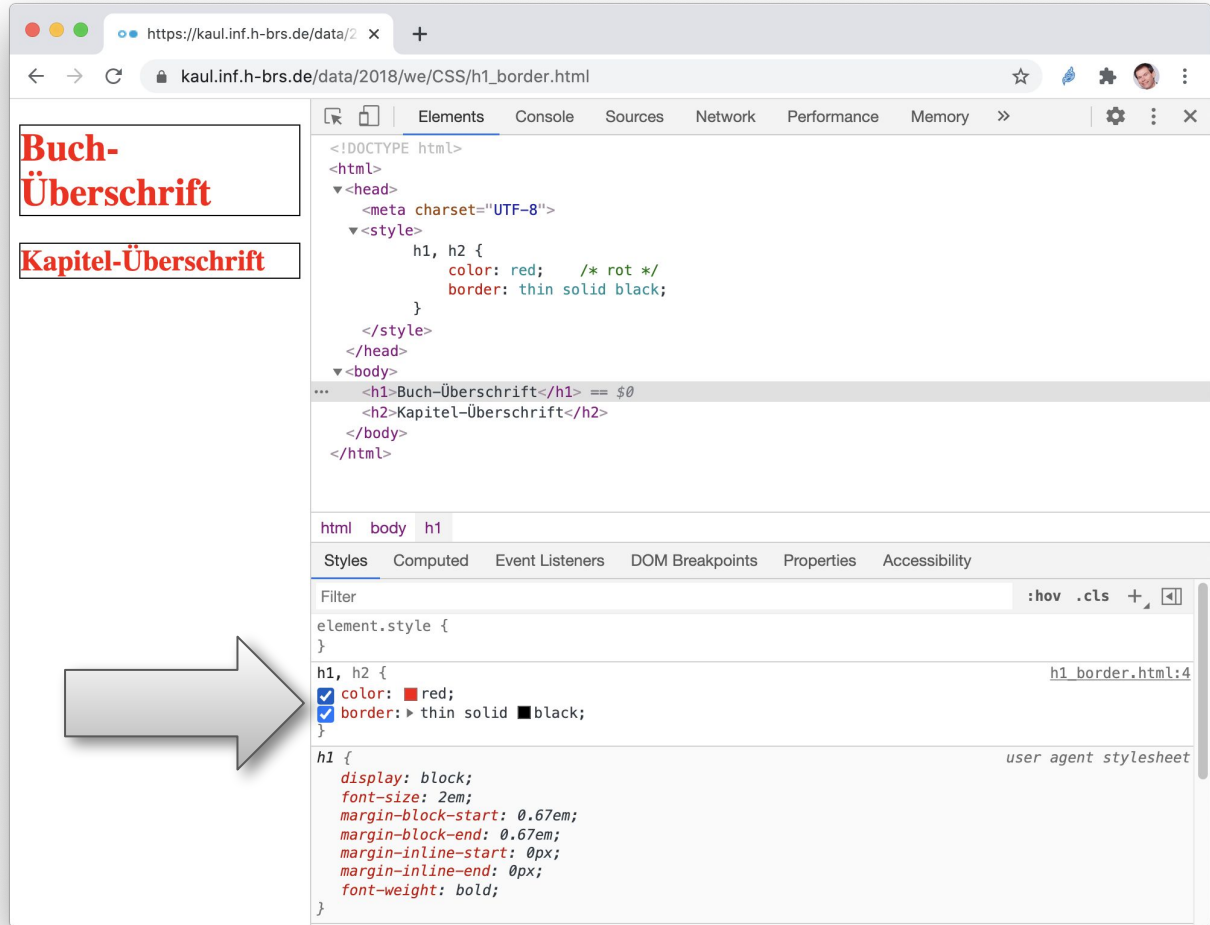
Beispiel:

```
h1, h2 {  
  color: red; /* rot */  
  border: thin solid black;  
}
```

Grouped Selectors

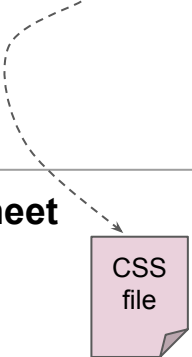
Grouped Declaration

Ansicht der CSS-Regeln in DevTools



3 Arten der Einbettung CSS in HTML

1	2	3
<link>	<style>	<tag>
<pre><!DOCTYPE html> <head> <title> Demo </title> <link rel="stylesheet" href="s.css"> </head> <h1> Header </h1></pre>	<pre><!DOCTYPE html> <style> h1 { color: red; } </style> <h1> Header </h1></pre>	<pre><!DOCTYPE html> <h1 style="color:red;"> Header </h1></pre>
external style sheet	internal style sheet	inline style



CSS file

CSS External Style Sheet

HTML-Datei

```
<!DOCTYPE html>
<html>
  <head>
    <title>Titel wird in Kopfzeile angezeigt</title>
    <meta charset="utf-8">
    <link rel="stylesheet" href="stylesheet.css">
  </head>
  <body>
    <h1>Überschrift</h1>
    
    <p> Inhalt eines Paragraphen mit
      <a href="seite2.html">Hyperlink</a>
    </p>
    <ul>
      <li>erster Listeneintrag</li>
      <li id="second">zweiter Listeneintrag</li>
    </ul>
  </body>
</html>
```

CSS-Datei

```
body {
  background-color: lightblue;
}

h1 {
  color: red;
  text-align: center;
}

a {
  background-color: white;
}

img.center {
  display: block;
  margin-left: auto;
  margin-right: auto;
}

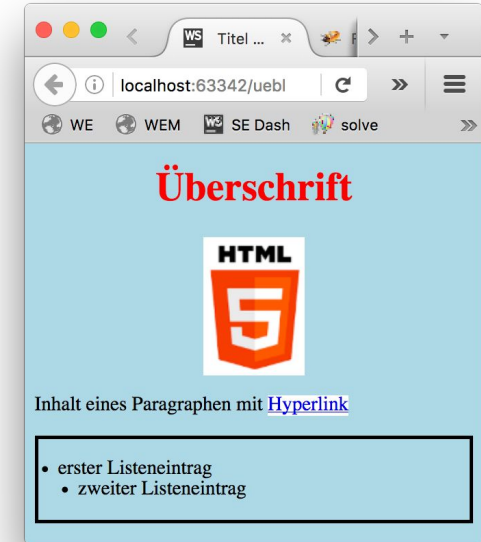
ul {
  border: solid;
  padding: 1em;
}

li#second {
  margin-left: 1em;
}
```

CSS-Syntax:

```
selector {
  property: value;
}
```

Browser



Selektoren stellen die Beziehung zwischen HTML und CSS her

Selektoren	HTML	CSS
Typ	<code><p></code>	<code>p { regel: wert; }</code>
Identifikator	<code><p id="eindeutig"></code>	<code>#eindeutig { color: red }</code>
Klasse	<code><p class="klasse"></code>	<code>.klasse</code>
Attribute	<code></code>	<code>[width="100"]</code>
Pseudoklassen	<code><div><p><p></div></code>	<code>:first-child { font-weight: bold; }</code> <code>:hover { color: red; }</code>
Pseudoelemente	<code><p>Am Anfang ...</p></code>	<code>::first-letter { font-size: large; }</code>

CSS-Selektoren (*eine Auswahl*)

Selektor	Selektion: alle Elemente mit ...
h1	Tag <h1>
<i>tag</i>	Tag < <i>tag</i> >
.myClass	Klasse myClass
#myId	ID myId
[foo]	Attribut foo
[foo=bar]	Attribut foo hat Value bar
[foo~=bar]	foo enthält bar
[foo^=bar]	foo beginnt mit bar
[foo\$=bar]	foo endet mit bar
[foo*=bar]	foo enthält bar

https://de.wikipedia.org/wiki/Cascading_Style_Sheets#Selektoren

Kombinatoren

Kombi	Selektion: alle Elemente
E F	alle Elemente F, die Nachfahren eines Elements E sind
E > F	alle Elemente F, die ein Kind eines Elements E sind
E ~ F	alle Elemente F, die einen Vorgänger E auf gleicher Ebene haben
E + F	alle Elemente F, die einen direkten Vorgänger E auf gleicher Ebene haben

<https://www.w3.org/TR/2018/CR-selectors-3-20180130/>

CSS-Selektoren (*Fortsetzung*)

Pseudoklassen

Selektor	Selektion: alle Elemente mit ...
:link	noch nicht besuchte Links
:visited	bereits besuchte Links
:active	das aktiv gerade angeklickte Element
:hover	Cursor über Element
:focus	Element im Fokus
:enabled	auswählbar
:checked	ausgewählt
:not(x)	x trifft nicht zu

strukturelle Pseudoklassen

Selektor	Selektion: alle Elemente mit ...
:first-child	erstes Kind
:last-child	letztes Kind
:nth-child(3)	drittes Kind
:nth-last-child(3)	dritt-letztes Kind

Vollständige Liste aller CSS-Selektoren Level 3

einfache Selektoren

- Universalselektor: *
- Typselektoren: element
- Klassenselektoren: .class
- ID-Selektoren: #id
- Attributselektoren
 - Attributpräsenz: [a]
 - Attributwert: [a=v], [a~=v], [a|=v]
 - Teilübereinstimmung: [a^=v], [a\$=v], [a*=v]

Kombinatoren

- Kindselektor: e > f
- Nachfahrenselektor: e f
- Nachbarselektor: e + f
 - The lobotomized owl selector: * + *
- Geschwisterselektor: e ~ f

Pseudoelemente

- ::first-line
- ::first-letter
- ::before, ::after
- ::backdrop
- ::selection

Pseudoklassen

• strukturelle Pseudoklassen

- :root
- :empty
- :first-child
- :last-child
- :nth-child()
- :nth-last-child()
- :only-child
- :first-of-type
- :last-of-type
- :nth-of-type()
- :nth-last-of-type()
- :only-of-type

• dynamische Pseudoklassen

- :any-link, :link, :visited
- :hover, :active, :focus
- :target
- :disabled, :enabled, :checked
- :valid, :invalid
- :in-range, :out-of-range



- :lang()
- :not()
- :matches()

CSS-Selektoren Level 4

Selectors Level 4

Editor's Draft, 21 March 2018

<https://drafts.csswg.org/selectors-4/>

```
/* The following selector matches all p elements immediately followed by an img element. */  
p:has(+img) {  
  color: red;  
}  
  
/* The following selector matches all section elements which contain h1, h2, and/or h3 elements. */  
section:has(h1, h2, h3) {  
  background-color: green;  
}  
  
/* The following selector matches anything containing a span element with a class urgent. */  
:has(span.urgent) {  
  background-color: red;  
}
```



caniuse css has

:has() CSS relational pseudo-class 📖 - WD

Usage % of all users
Global 0%

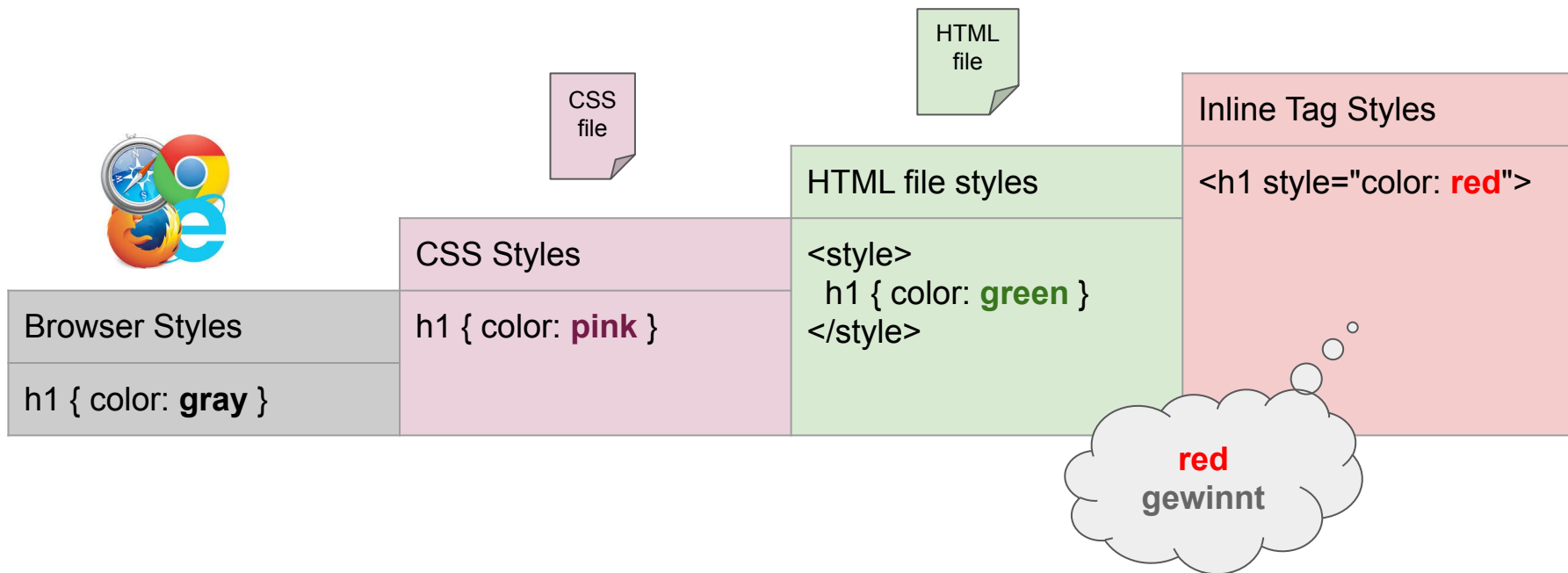
Only select elements containing specified content. For example,
`a:has(>img)` selects all `<a>` elements that contain an `<img>` child.

Current aligned Usage relative Date relative Show all

IE	Edge *	Firefox	Chrome	Safari	iOS Safari *	Opera Mini *	Chrome for Android	UC Browser for Android	Samsung Internet
			49						
			63		10.3				
		58	64	11	11.2				4
11	16	59	65	11.1	11.3	all	64	11.8	6.2
	17	60	66	TP					
	18	61	67						
			68						

Die CSS-Kaskade

- Styles der höheren Ebene überschreiben Styles darunter liegenden Ebenen



Die CSS-Kaskadierungsregeln

- Die letzte Deklaration gewinnt.
- Inline-Styles > Webseiten-Styles > externe CSS-Dateien > Browser
- Ablauf der Kaskade (mit steigender Wichtigkeit):
 - Deklarationen im Browser-Stylesheet
 - Deklarationen des Benutzers
 - Deklarationen des Autors
 - Wichtige Deklarationen (mit !important) des Autors
 - Wichtige Deklarationen (mit !important) des Benutzers

<http://wiki.selfhtml.org/wiki/CSS/Kaskade>

<https://stackoverflow.com/questions/37910944/set-multiple-values-for-one-css-attribute>

Spezifität = Prioritätsregeln in CSS

Je spezifischer, desto höher die Priorität

- A = ID-Selektor
- B = Anzahl der Attribut-, Klassenselektoren und Pseudoklassen
- C = Anzahl der Typselektoren und Pseudoelemente
- **!important** höchste Prio

Beispiel: - Spezifität von Selektoren von gering nach hoch geordnet

*	/* A=0, B=0, C=0, Spezifität 0 0 0 */
h1	/* A=0, B=0, C=1, Spezifität 0 0 1 */
ul li	/* A=0, B=0, C=2, Spezifität 0 0 2 */
a::after	/* A=0, B=0, C=2, Spezifität 0 0 2 */
p:first-child	/* A=0, B=1, C=1, Spezifität 0 1 1 */
a:not([href])	/* A=0, B=1, C=1, Spezifität 0 1 1 */
ul.nav [href]	/* A=0, B=2, C=1, Spezifität 0 2 1 */
#author	/* A=1, B=0, C=0, Spezifität 1 0 0 */
#editor p	/* A=1, B=0, C=1, Spezifität 1 0 1 */

```
h1 { color: green !important; }  
h1 { color: red; }
```

<https://wiki.selfhtml.org/wiki/Spezifität>

`body #content .alternative p` has a specificity of **112** (HTML selector + id selector + class selector + HTML selector, 1+100+10+1)

<http://www.htmldog.com/guides/css/intermediate/specificity/>

CSS Specificity Calculator

Specificity Calculator

A visual way to understand [CSS specificity](https://specificity.keegan.st/). Change the selectors or paste in your own.

`li:first-child h2 .title`

0

Inline styles

0

IDs

2

Classes, attributes
and pseudo-classes

2

Elements and
pseudo-elements

+ Duplicate

`#nav .selected > a:hover`

0

Inline styles

1

IDs

2

Classes, attributes
and pseudo-classes

1

Elements and
pseudo-elements

+ Duplicate

<https://specificity.keegan.st/>

Aufgaben zur Spezifität von CSS-Selektoren

Selektor	Spezifität
body #content .alternative p	
li:first-child h2 .title	
ul ul li nth-child(2)	
div > div > ul > li + li	
section > .my > ul > li .li	

CSS kennt 3 Farbnotationen

- 1. Schlüsselworte

- black, white, blue, green, ...

4.4 Recognized color keyword names			
The following is the list of recognized color keywords that can be used as a keyword value for data type <code><color></code> :			
 <code>aliceblue</code>	<code>rgb(240, 248, 255)</code>	 <code>lightpink</code>	<code>rgb(255, 182, 193)</code>
 <code>antiquewhite</code>	<code>rgb(250, 235, 215)</code>	 <code>lightsalmon</code>	<code>rgb(255, 160, 122)</code>
 <code>aqua</code>	<code>rgb(0, 255, 255)</code>	 <code>lightseagreen</code>	<code>rgb(32, 178, 170)</code>

- 2. RGB-Farbraum

- Hexadezimale Notation `#RRGGBB` und `#RGB`, z.B. `#FF00A0`
 - Funktionale RGB-Notation `rgb(R,G,B)`, z.B. `rgb(255, 0, 51)`

- 3. HSL zylindrisches Koordinatensystem

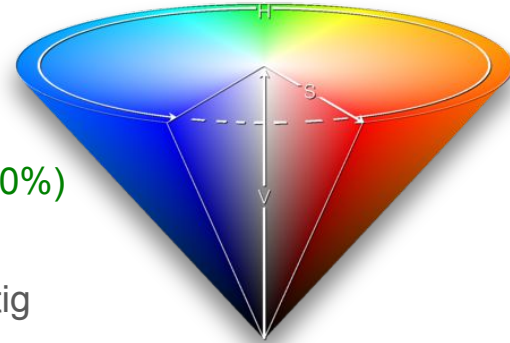
- Hue-Saturation-Lightness-Notation `hsl(H,S,L)`, z.B. `hsl(120,100%,50%)`

- zusätzlich Alpha-Kanal

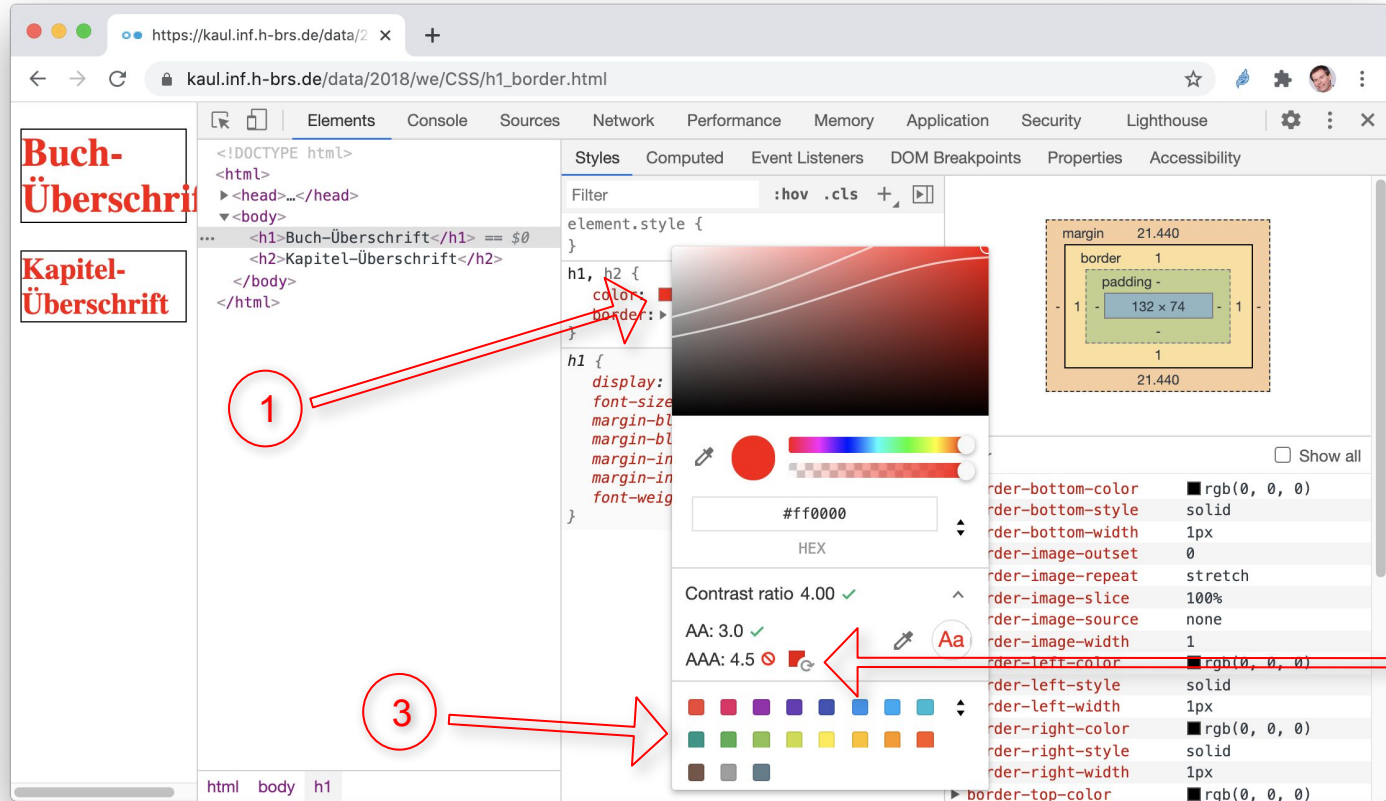
- `rgba()` mit Alpha-Kanal = Deckkraft: 0=durchsichtig; 1=undurchsichtig
 - `hsla()` = `hsl()` + Alpha-Kanal

- Schlüsselworte

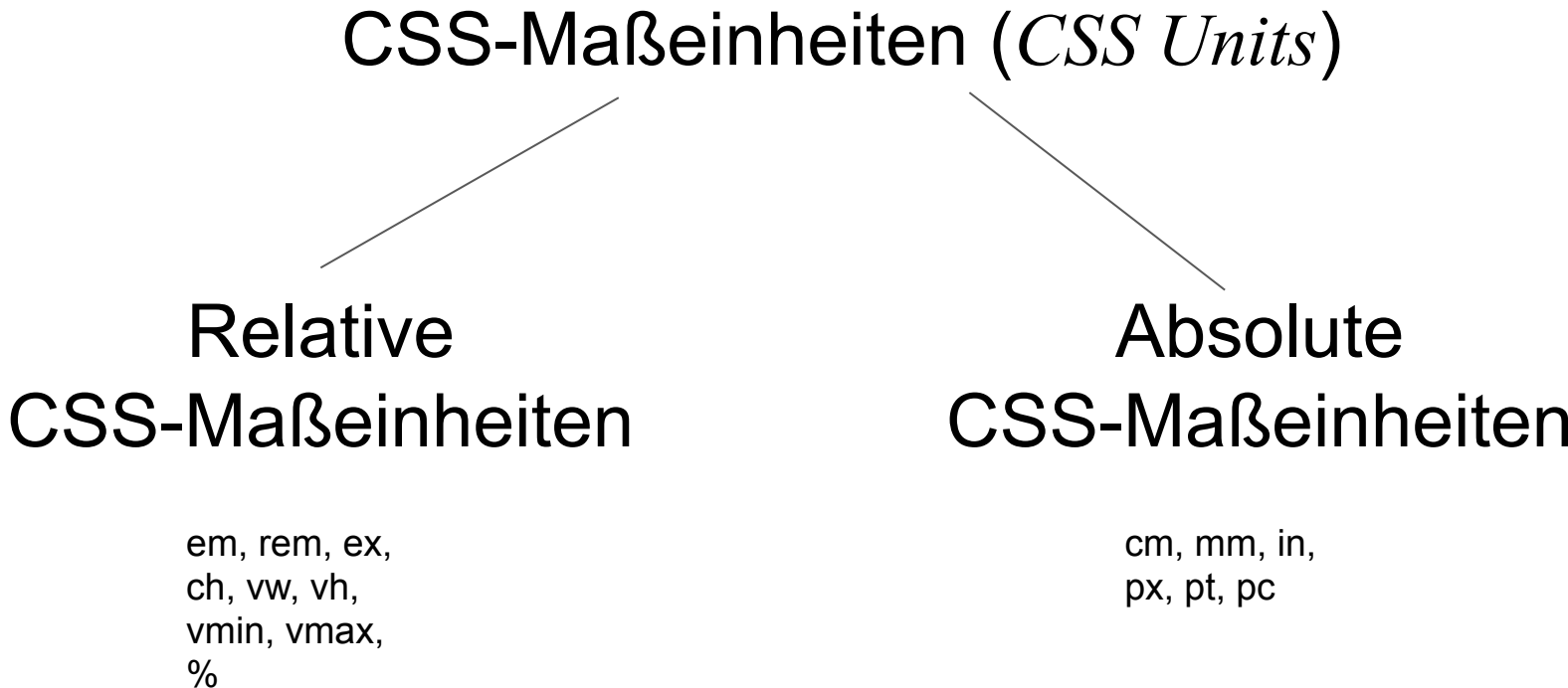
- `currentColor`
 - `transparent` = `rgba(0,0,0,0)` = "nicht deckendes Weiß"



DevTools Color Picker & Color Contrast Checker



CSS-Maßeinheiten (*CSS Units*)



Relative CSS-Maßeinheiten

em, rem, ex,
ch, vw, vh,
vmin, vmax,
%

Absolute CSS-Maßeinheiten

cm, mm, in,
px, pt, pc

<https://www.w3.org/TR/css-values-3/>

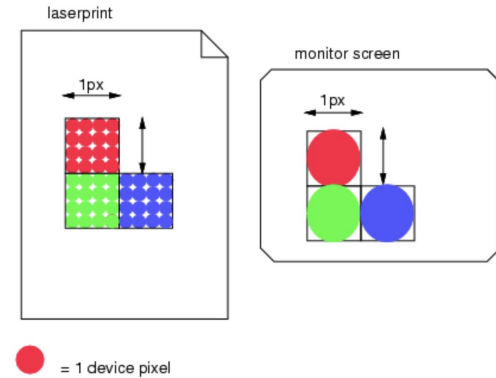
Relative CSS-Maßeinheiten (*Relative CSS Units*)

Unit	Description
em	Relative to the font-size of the element (2em means 2 times the size of the current font)
ch	Relative to width of the "0" (zero)
rem	Relative to font-size of the root element
vw	Relative to 1% of the width of the viewport
vh	Relative to 1% of the height of the viewport
vmin	Relative to 1% of viewport's smaller dimension
vmax	Relative to 1% of viewport's larger dimension
%	Prozent

https://www.w3schools.com/cssref/css_units.asp

Absolute CSS-Maßeinheiten (*Absolute CSS Units*)

Einheit	Beschreibung
cm	Zentimeter
mm	Millimeter
in	Inches (1in = 96px = 2.54cm)
px	Pixels (1px = 1/96th of 1in) $\approx$ (früher) ein Bildschirm-Pixel
pt	Points (1pt = 1/72 of 1in)
pc	Picas (1pc = 12 pt)



<https://www.w3.org/TR/css-values-3/#absolute-lengths>

https://www.w3schools.com/cssref/css_units.asp

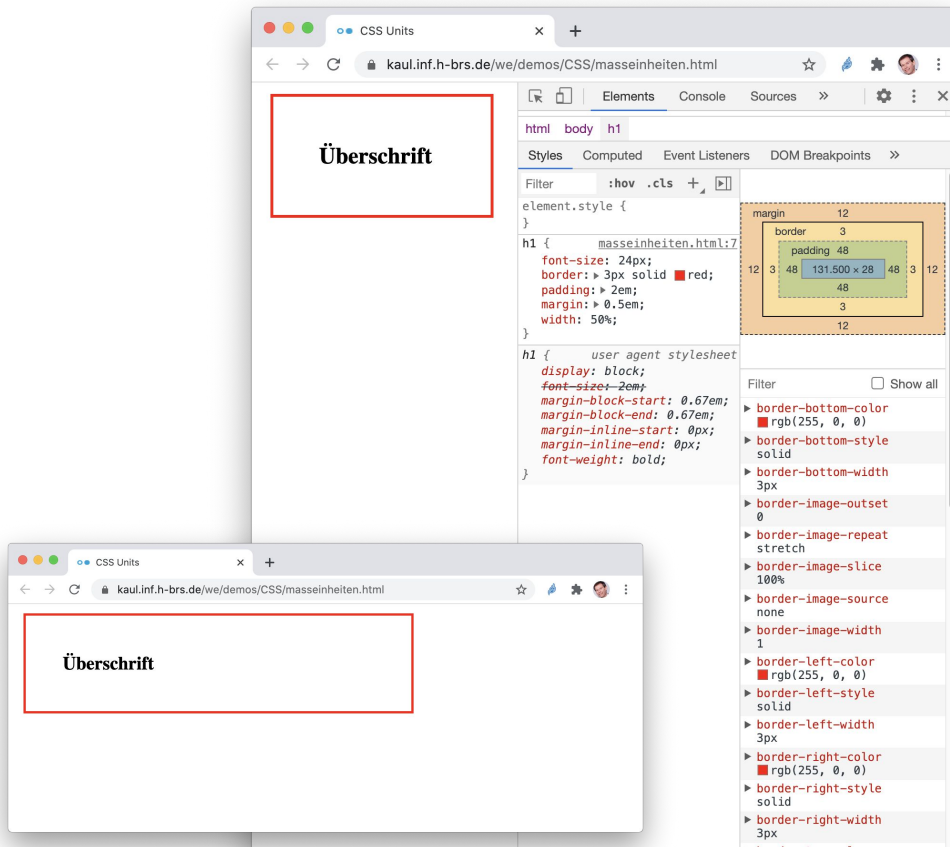
CSS-Maßeinheiten für Winkel

deg	Ein Vollwinkel, also eine Kreisumrundung, entspricht $360^\circ = 360\text{deg}$
grad	100 Gon entsprechen einer 90° -Drehung, also einem rechten Winkel. Eine Kreisumrundung entspricht somit 400 Gon, also $360^\circ = 400\text{grad}$.
rad	Ein Radiant entspricht einer Drehung, bei dem der Winkel durch die Länge des entsprechenden Kreisbogens im Einheitskreis angegeben wird. Eine Kreisumrundung entspricht somit $360^\circ = 2\pi \text{ rad}$.
turn	Ein Vollwinkel entspricht einer Kreisumrundung, d.h. $360^\circ = 1\text{turn}$

<https://www.w3.org/TR/css-values-3/#angles>

Beispiel CSS Maßeinheiten (*CSS Units*)

```
<!DOCTYPE html>
<html lang="de">
<head>
  <meta charset="UTF-8">
  <title>CSS Units</title>
  <style>
    h1 {
      font-size: 24px;
      border: 3px solid red;
      padding: 2em;
      margin: 0.5em;
      width: 50%;
    }
  </style>
</head>
<body>
  <h1>Überschrift</h1>
</body>
</html>
```



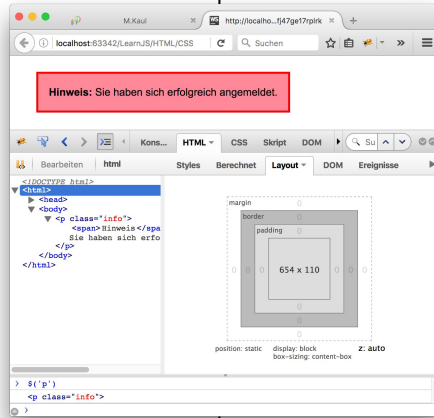
Das CSS-Box-Modell

Regel: In HTML sind alle Einheiten (Boxen) rechteckig.

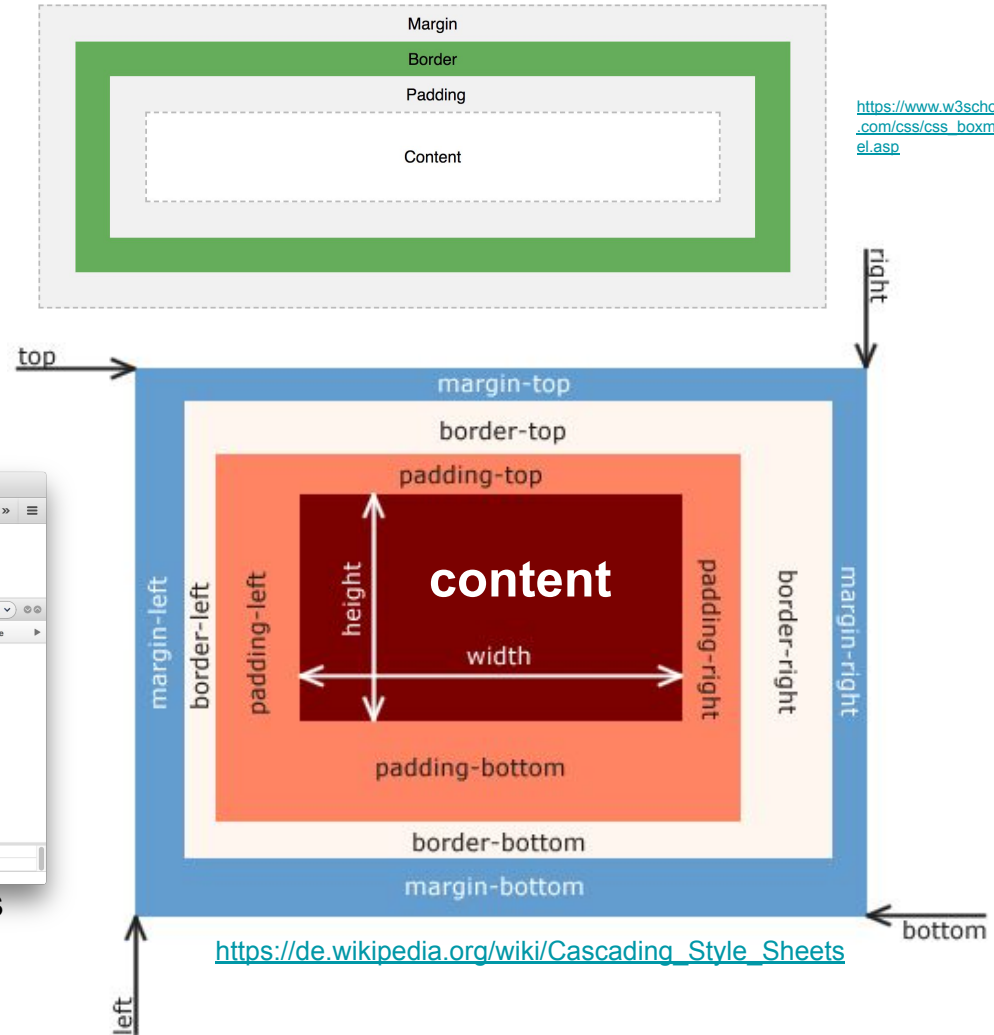
```
<!DOCTYPE html>
<style>
  p.info {
    font-family: arial, sans-serif;
    line-height: 150%;
    margin-left: 2em;
    padding: 1em;
    border: 3px solid red;
    background-color: #f89;
    display: inline-block;
  }
```

```
p.info span {
  font-weight: bold;
}
p.info span::after {
  content: ":", " ";
}
</style>
```

```
<p class="info">
  <span>Hinweis</span>
  Sie haben sich erfolgreich angemeldet.
</p>
```



DevTools



Umschalten zwischen Box-Bemaßungsmodellen

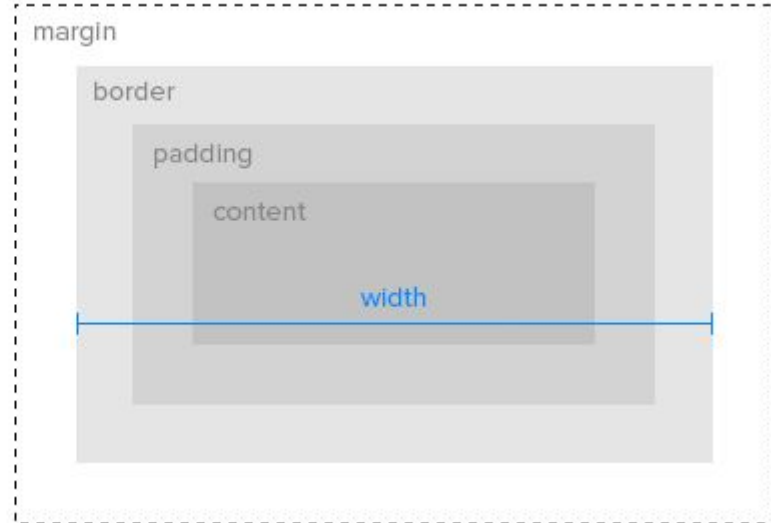
`box-sizing: content-box;`

W3C box model



`box-sizing: border-box;`

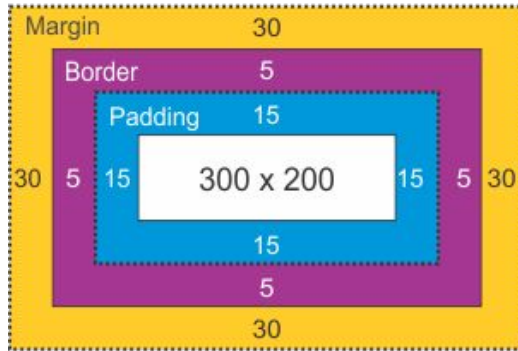
Internet Explorer box model



Umschalten zwischen Box-Modellen

box-sizing: content-box;

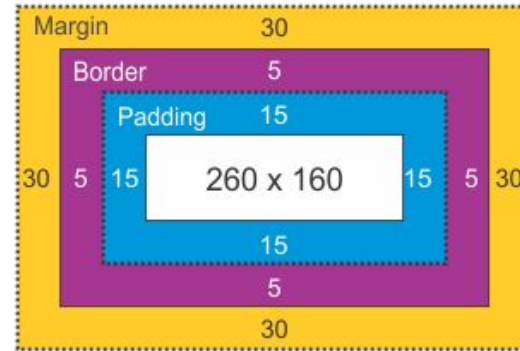
Box Model is content-box



```
div{  
  width: 300px;  
  height: 200px;  
  padding: 15px;  
  border: 5px solid grey;  
  margin: 30px;  
  -moz-box-sizing: content-box;  
  -webkit-box-sizing: content-box;  
  box-sizing: content-box;  
}
```

box-sizing: border-box;

Box Model is border-box



```
div{  
  width: 300px;  
  height: 200px;  
  padding: 15px;  
  border: 5px solid grey;  
  margin: 30px;  
  -moz-box-sizing: border-box;  
  -webkit-box-sizing: border-box;  
  box-sizing: border-box;  
}
```

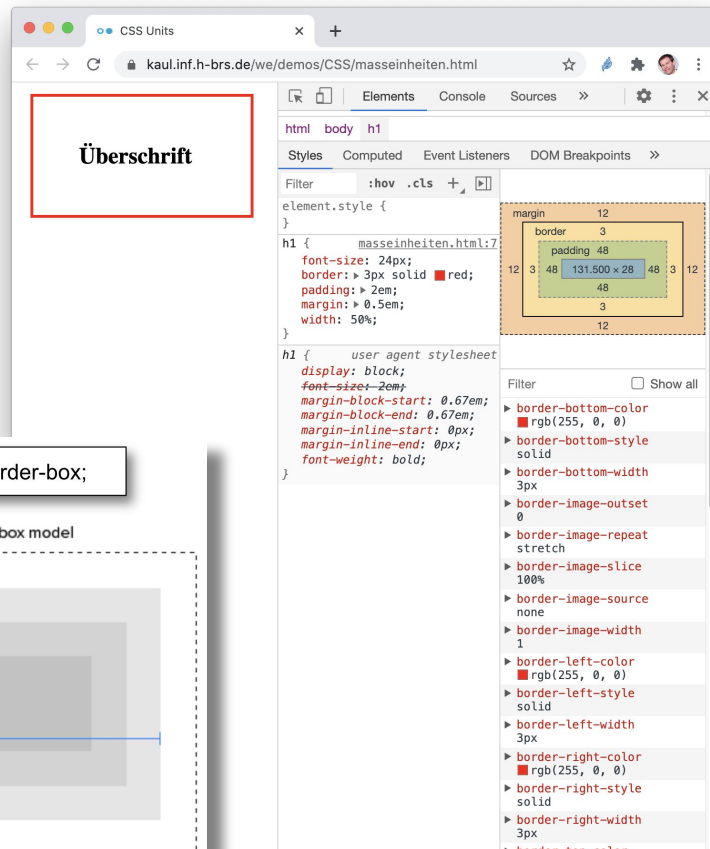
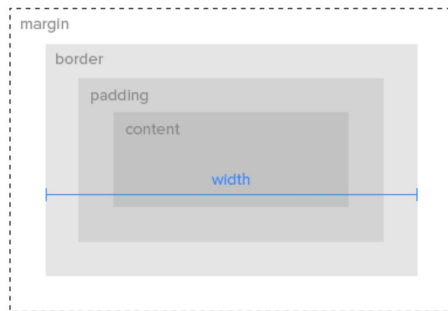
Lösung mit `box-sizing: border-box;`

```
<!DOCTYPE html>
<html lang="de">
<head>
  <meta charset="UTF-8">
  <title>CSS Units</title>
  <style>
    h1 {
      box-sizing: border-box;
      font-size: 24px;
      border: 3px solid red;
      padding: 1em;
      margin: 0.5em;
      width: calc( 100% - 1em )
    }
  </style>
</head>
<body>
  <h1>Überschrift</h1>
</body>
</html>
```



box-sizing: border-box;

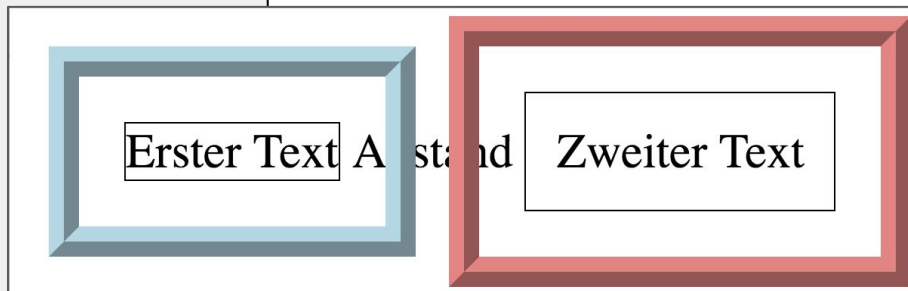
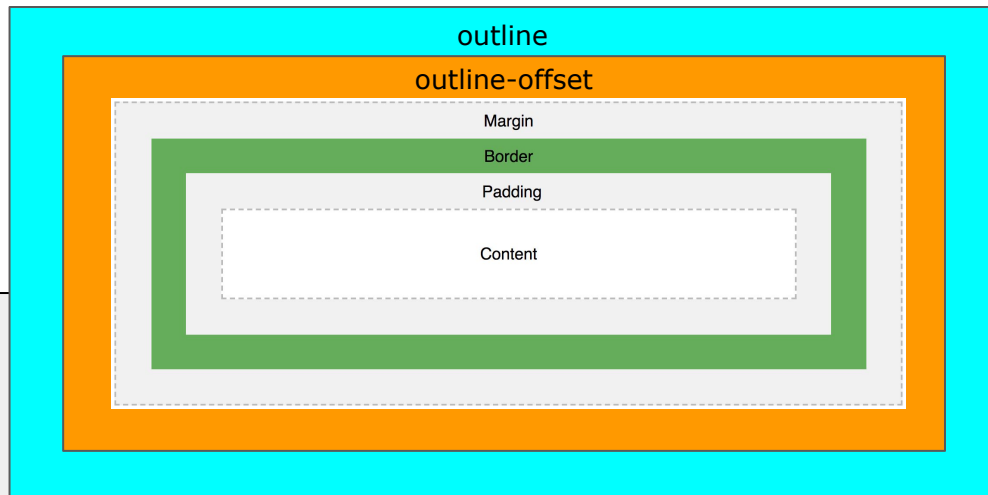
Internet Explorer box model



Outline

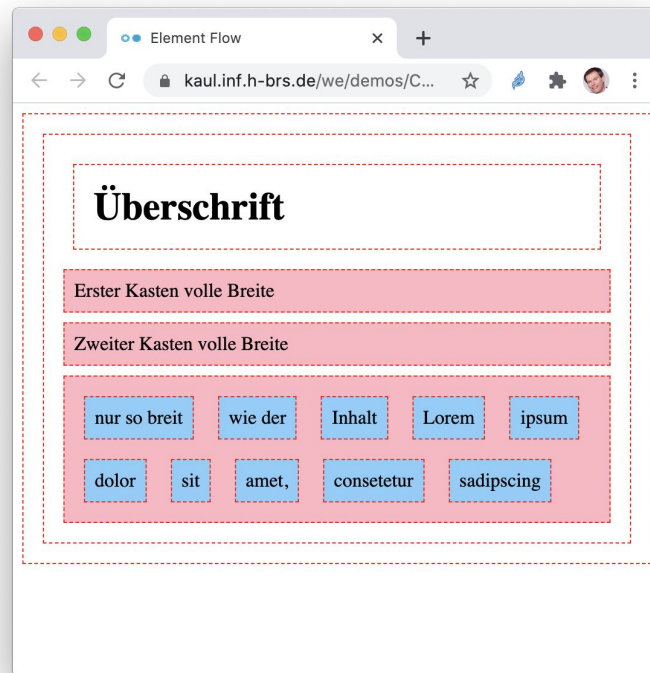
Noch ein Rahmen außerhalb der Box, der Nachbarboxen überdecken kann

```
<style>
body { padding: 30px; }
#a {
  border: thin solid black;
  outline: 10px ridge lightblue;
  outline-offset: 15px;
}
#b {
  padding: 10px;
  border: thin solid black;
  outline: 10px ridge lightcoral;
  outline-offset: 15px;
}
</style>
<body>
  <span id="a">Erster Text</span> Abstand
  <span id="b">Zweiter Text</span>
</body>
```



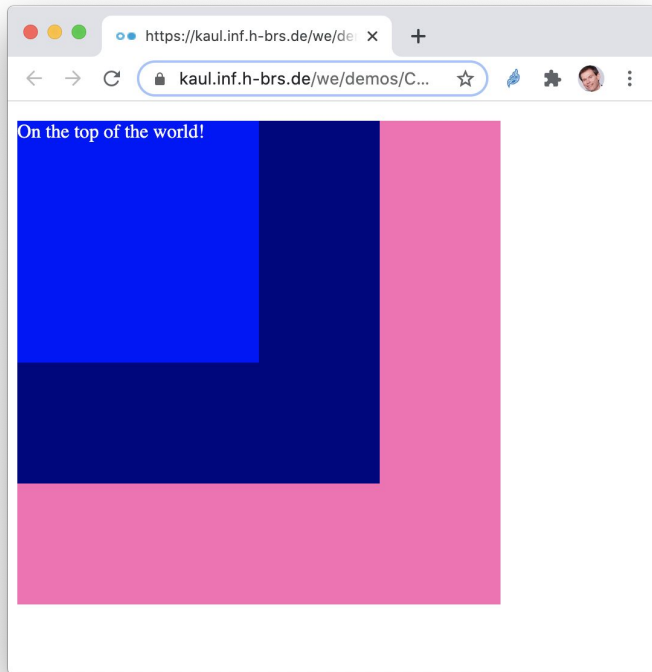
Elementfluss für Block- und Inline-Elemente

```
<!DOCTYPE html>
<head>
  <style>
    * {
      border: thin dashed red;
      padding: 0.5em;
      margin: 0.5em;
    }
    div {
      background-color: lightpink;
    }
    span {
      background-color: lightskyblue;
      display: inline-block;
    }
  </style>
</head>
<body>
  <h1>Überschrift</h1>
  <div>Erster Kasten volle Breite
</div>
  <div>Zweiter Kasten volle Breite
</div>
  <div>
    <span>nur so breit</span> <span>wie der </span> <span>Inhalt</span>
    <span>Lorem</span> <span>ipsum</span> <span>dolor</span> <span>sit
</span> <span>amet,</span> <span>consetetur</span> <span>sadipscing</span>
  </div>
</body>
</html>
```



Stapel von Block-Elementen übereinander

```
<!DOCTYPE html>
<style>
  .grandpa {
    width: 400px;
    height: 400px;
    background-color: hotpink;
  }
  .papa {
    width: 300px;
    height: 300px;
    background-color: navy;
  }
  .son {
    width: 200px;
    height: 200px;
    background-color: blue;
  }
  p{
    color: white;
  }
</style>
<div class="grandpa">
  <div class="papa">
    <div class="son">
      <p>On the top of the world!</p>
    </div>
  </div>
</div>
```



Positionierung

```
#id1 {  
  position: relative;  
  top: 20px;  
  left: -40px;  
}
```

```
#id2 {  
  position: absolute;  
  top: 0;  
  right: 0;  
  width: 200px;  
}
```

```
#id3 {  
  position: fixed;  
  bottom: 0;  
  right: 0;  
}
```

position: static	Default: normale Positionierung im Flow der Elemente
position: relative	Verschiebung des Elementes relativ zur normalen Position
position: absolute	Positionierung des Elementes relativ zum nächsten non-static positionierten Elternelement. Wenn es keins gibt, dann ViewPort.
position: fixed	Positionierung des Elementes relativ zum ViewPort (Tab-Fenster, Browser-Fenster)
position: sticky	Beibehalten der Position im Elementfluss, bis das obere oder untere Seitenende erreicht ist. Dann dort „kleben“ bleiben.

removed from element flow

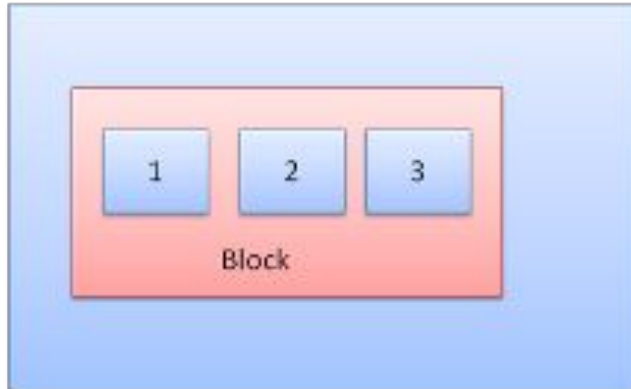


<https://css-tricks.com/absolute-relative-fixed-positioning-how-do-they-differ/>
<http://www.barelyfitz.com/screencast/html-training/css/positioning/>
https://www.w3schools.com/css/css_positioning.asp

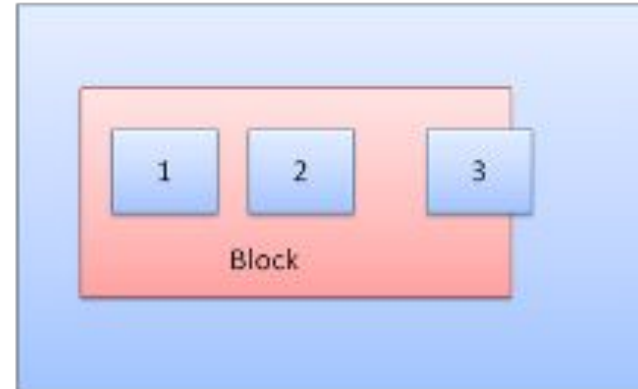
Position: Relative

```
<html>  
  <div>  
    <span>1</span>  
    <span>2</span>  
    <span style="position:relative;left:5px">3</span>  
  </div>  
</html>
```

Normal layout



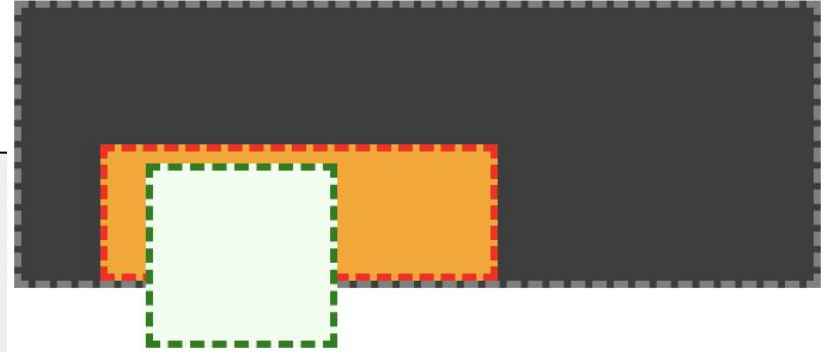
Applying relative positioning



Prozentangaben relativ zum Parent

```
<style>
* {
  box-sizing: border-box;
}
#outer {
  position: relative;
  height: 10em;
  background: #3e3e3e;
  border: dashed grey;
}
#inner {
  left: 10%;
  background: orange;
  border: dashed red;
  top: 50%;
  width: 50%;
  min-height: 50%;
  position: absolute;
}
#inner-inner {
  background: #effef;
  border: dashed green;
  position: absolute;
  width: 50%;
  height: 150%;
  top: 10%;
  left: 10%;
}
</style>
```

```
<div id="outer">
  <div id="inner">
    <div id="inner-inner"></div>
  </div>
</div>
```



CSS *display*-Property: display: inline;

Value	Description
inline	Default value. Displays an element as an inline element (like)
block	Displays an element as a block element (like <p>)
flex	Displays an element as a block-level flex container. New in CSS3
grid	Displays an element as a block-level grid container. New in CSS3
inline-block	Displays an element as an inline-level block container. The inside of this block is formatted as block-level box, and the element itself is formatted as an inline-level box
inline-flex	Displays an element as an inline-level flex container. New in CSS3
inline-table	The element is displayed as an inline-level table
list-item	Let the element behave like a element
run-in	Displays an element as either block or inline, depending on context
none	The element will not be displayed at all (has no effect on layout)
initial	Sets this property to its default value.
inherit	Inherits this property from its parent element.

Die *display*-Property -2-

display: table

table	Let the element behave like a <table> element
table-caption	Let the element behave like a <caption> element
table-column-group	Let the element behave like a <colgroup> element
table-header-group	Let the element behave like a <thead> element
table-footer-group	Let the element behave like a <tfoot> element
table-row-group	Let the element behave like a <tbody> element
table-cell	Let the element behave like a <td> element
table-column	Let the element behave like a <col> element
table-row	Let the element behave like a <tr> element

CSS Layout

Ein [CSS Layoutmodus](#), manchmal als *Layout* abgekürzt, ist ein **Algorithmus**, der die **Position und die Größe von Boxen** bestimmt, basierend auf der Weise, wie diese mit ihren Geschwisterknoten und übergeordneten Knoten **interagieren**. Es gibt mehrere davon:

1. Das *Blocklayout*, designt, um Dokumente darzustellen. Das Blocklayout beinhaltet dokumentenspezifische Features, wie die Möglichkeit, Elemente [umzubrechen](#) oder sie in [mehreren Spalten](#) darzustellen.
2. Das *Inlinelayout*, designt, um Text darzustellen.
3. Das *Tabellenlayout*, designt, um Tabellen darzustellen.
4. Das *positionierte Layout*, designt, um Elemente ohne viel Interaktion mit anderen Elementen zu positionieren.
5. Das [Flexboxlayout](#), designt, um komplexe Seiten darzustellen, die problemlos in der Größe geändert werden können.
6. Das *Rasterlayout (oder Gridlayout)*, designt, um Elemente relativ zu einem festen Raster darzustellen.

Welcher Layoutmodus zum Tragen kommt, wird mit `display` bestimmt.

Block und Inline Layout

```
graph TD; A[Block und Inline Layout] --- B[display: block;]; A --- C[display: inline;]; A -.- D[Inline-Block Layout];
```

display: block;

wie <h1>, <p>, <div>

mit width, height, margin

display: inline;

wie <a>, , ,

ohne width, height, margin

Inline-Block Layout

display: inline-block;

mit width, height, margin

Inline-Block Layout

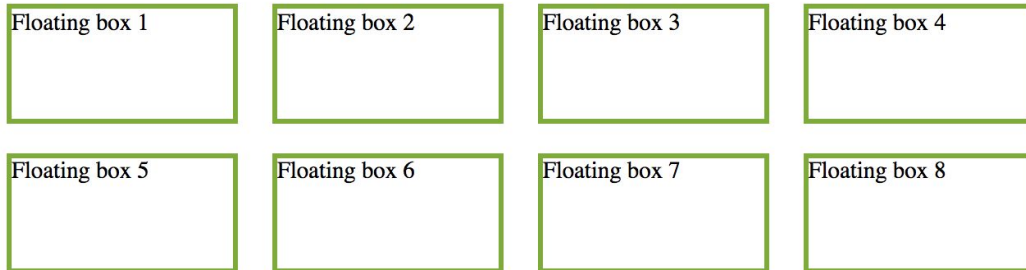
fließen wie Inline-Elemente, jedoch mit Höhe und Breite

```
<style>
.floating-box {
  display: inline-block;
  width: 150px;
  height: 75px;
  margin: 10px;
  border: 3px solid #73AD21;
}
.after-box {
  border: 3px solid red;
}
</style>
```

```
<div class="floating-box">Floating box 1</div>
<div class="floating-box">Floating box 2</div>
<div class="floating-box">Floating box 3</div>
<div class="floating-box">Floating box 4</div>
<div class="floating-box">Floating box 5</div>
<div class="floating-box">Floating box 6</div>
<div class="floating-box">Floating box 7</div>
<div class="floating-box">Floating box 8</div>

<div class="after-box">
  Another box, after the floating boxes...</div>
```

The New Way - using inline-block



https://www.w3schools.com/css/css_inline-block.asp

Float Layout

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[CSS How To](#)
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[CSS Margins](#)
[CSS Padding](#)
[CSS Height/Width](#)
[CSS Box Model](#)
[CSS Outline](#)
[CSS Text](#)
[CSS Fonts](#)
[CSS Icons](#)
[CSS Links](#)
[CSS Lists](#)
[CSS Tables](#)
[CSS Display](#)
[CSS Max-width](#)
[CSS Position](#)
[CSS Overflow](#)
[CSS Float](#)

Example - float: right;

The following example specifies that an image should float to the **right** in a text:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus imperdiet, nulla et dictum interdum, nisi lorem egestas odio, vitae scelerisque enim ligula venenatis dolor. Maecenas nisl est, ultrices nec congue eget, auctor vitae massa. Fusce luctus vestibulum augue ut aliquet. Mauris ante ligula, facilisis sed ornare eu, lobortis in odio. Praesent convallis urna a lacus interdum ut hendrerit risus congue. Nunc sagittis dictum nisi, sed ullamcorper ipsum dignissim ac...



Example

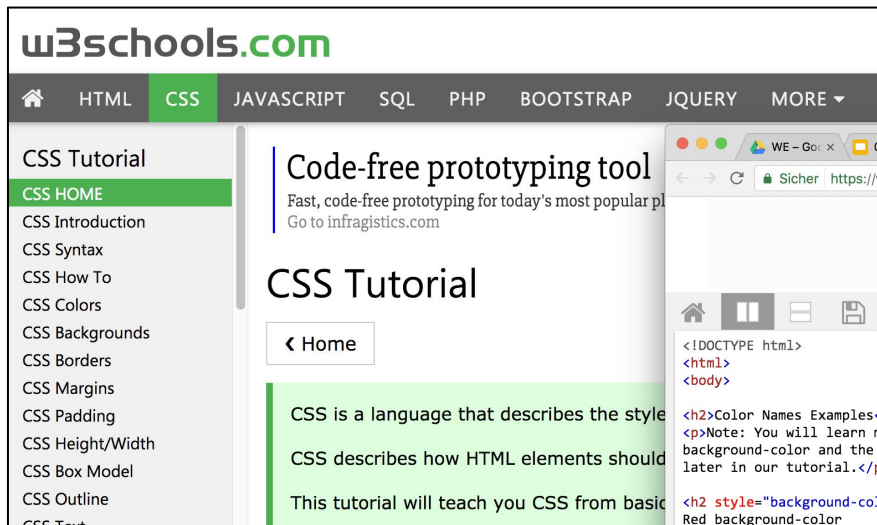
```
img {  
  float: right;  
}
```

Try it Yourself »

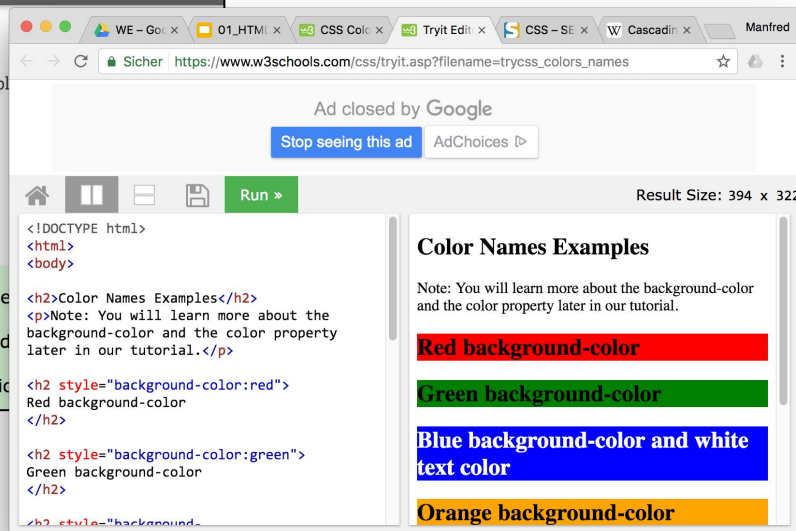
Wie lernt man CSS?



<https://wiki.selfhtml.org/wiki/CSS>



TryIt-Editor



<https://www.w3schools.com/css/default.asp>

weitere Tutorial-Angebote am Ende ...

W3Schools.com

https://www.w3schools.com/css/css_syntax.asp

WE_BCS - Google Dr x

02_CSS - Google Pr x

CSS Syntax and Sele x

CSS - SELFHTML-Wi x

CSS | MDN x

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Sicher https://www.w3schools.com/css/css_syntax.asp

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CSS Introduction

CSS Syntax

CSS How To

CSS Colors

CSS Backgrounds

CSS Borders

CSS Margins

CSS Padding

CSS Height/Width

CSS Box Model

CSS Outline

CSS Text

CSS Fonts

CSS Icons

CSS Links

CSS Lists

CSS Tables

CSS Display

CSS Max-width

CSS Position

CSS Overflow

CSS Float

CSS Inline-block

CSS Align

CSS Combinators

CSS Syntax and Selectors

Previous

CSS Syntax

A CSS rule-set consists of a selector and a declaration block:

Selector

Declaration

Declaration

h1

{ color:blue; font-size:12px; }

Property

Value

Property

Value

The selector points to the HTML element you want to style.

The declaration block contains one or more declarations separated by semicolons.

Each declaration includes a CSS property name and a value, separated by a colon.

A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

In the following example all <p> elements will be center-aligned, with a red text color:

Example

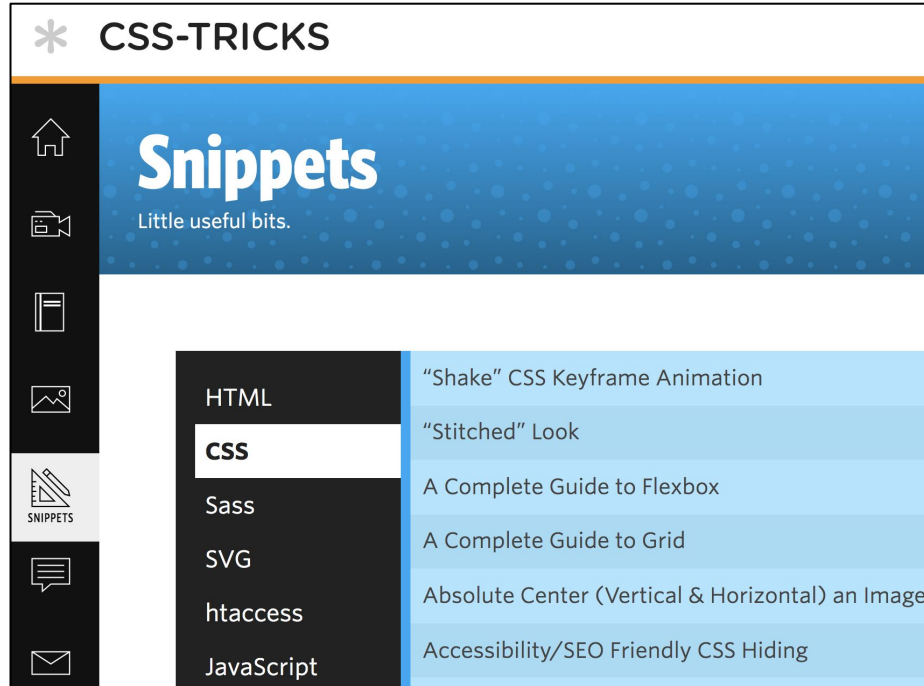
```
p {
  color: red;
  text-align: center;
}
```

CSS Tutorial
CSS HOME
CSS Introduction
CSS Syntax
CSS How To
CSS Colors
CSS Backgrounds
CSS Borders
CSS Margins
CSS Padding
CSS Height/Width
CSS Box Model
CSS Outline
CSS Text
CSS Fonts
CSS Icons
CSS Links
CSS Lists
CSS Tables
CSS Display
CSS Max-width
CSS Position
CSS Overflow
CSS Float
CSS Inline-block
CSS Align
CSS Combinators
CSS Pseudo-class
CSS Pseudo-element
CSS Opacity
CSS Navigation Bar
CSS Dropdowns
CSS Tooltips
CSS Image Gallery
CSS Image Sprites
CSS Attr Selectors
CSS Forms
CSS Counters

CSS3
CSS3 Introduction
CSS3 Rounded Corners
CSS3 Border Images
CSS3 Backgrounds
CSS3 Colors
CSS3 Gradients
CSS3 Shadows
CSS3 Text
CSS3 Fonts
CSS3 2D Transforms
CSS3 3D Transforms
CSS3 Transitions
CSS3 Animations
CSS3 Images
CSS3 Buttons
CSS3 Pagination
CSS3 Multiple Columns
CSS3 User Interface
CSS3 Box Sizing
CSS3 Flexbox
CSS3 Media Queries
CSS3 MQ Examples

CSS Responsive
RWD Intro
RWD Viewport
RWD Grid View
RWD Media Queries
RWD Images
RWD Videos
RWD Frameworks
RWD Templates
CSS Examples
CSS Examples
CSS Quiz
CSS Exercises
CSS Certificate
CSS References
CSS Reference
CSS Selectors
CSS Functions
CSS Reference Aural
CSS Web Safe Fonts
CSS Animatable
CSS Units
CSS PX-EM Converter
CSS Colors
CSS Color Values
CSS Default Values
CSS3 Browser Support

CSS Tricks

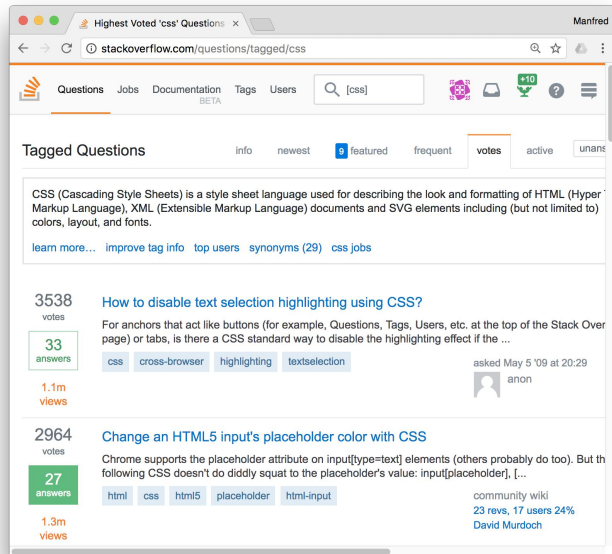


<https://css-tricks.com/>

Tools

- Lorem Ipsum Generator <http://www.loremipsum.de/>
- Responsive Font Size Caculator
<http://emilolsson.com/tools/vw-unit-calc-an-online-responsive-css-font-size-calculator/>
- CSS Specifity Calculator <https://specificity.keegan.st/>
- CSS Unit Conversion Calculator
<https://sabe.io/tools/css-unit-conversion-calculator>

Wo bekommt man Antworten?



<http://stackoverflow.com/questions/tagged/css>

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- CSS Flex Layout
- CSS Grid Layout
- CSS Rounded Corners
- CSS Border Images
- CSS Backgrounds
- CSS Colors
- CSS Gradients
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- CSS Text Effects
- CSS Web Fonts
- CSS 2D Transforms
- CSS 3D Transforms
- CSS Transitions
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<https://www.w3schools.com/css/>